

Limits:—• $(-\infty < x < \infty)$

- 1) Normal Distribution -
- 2) Cauchy Distribution -
- 3) standard cauchy Distribution -
- 4) In Bivariate Distribution -

 $pdf (x, y)$

- 5) t - Distribution -

• $(0 < x < \infty)$

- 1) Even function, Odd Function

Gamma function

- 2) Distribution function $(0 \rightarrow x)$.
- 3) lognormal Distribution
- 4) Exponential Distribution
- 5) Gamma Distribution
- 6) Beta Distribution (with 2nd kind)
- 7) Weibull Distribution
- 8) chi-square Distribution
- 9) F - Distribution

• $(-x < x < a) \rightarrow$ Median• $(0 \leq x \leq 1) \rightarrow$ Beta Distribution• $(K < x < \infty) \rightarrow$ paretto Distribution• $(-1 < x < 1) \rightarrow$ In Bivariate Distribution pdf

Functions:—(1) Normal Distribution:

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

(2) Cauchy Distribution:

$$f(x) = \frac{1}{\pi (\lambda^2 + (x-\mu)^2)} \cdot dx$$

(3) Standard cauchy distribution:

$$f(x) = \frac{1}{\pi (1+y^2)} \cdot dx$$

(4) Bivariate Distribution:

$$f(x,y) = \frac{1}{2\pi \sigma_1 \sigma_2 \sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x-\mu_1}{\sigma_1} \right) \left(\frac{y-\mu_2}{\sigma_2} \right) + \left(\frac{y-\mu_2}{\sigma_2} \right)^2 \right\} \right\}$$

$$-\infty < x < \infty$$

$$-\infty < y < \infty$$

$$-1 < \rho < 1$$

$$-\infty < \mu_1 < \infty$$

$$-\infty < \mu_2 < \infty$$

$$\sigma_1, \sigma_2 > 0$$

(5) t-Distribution:

$$t = \frac{x}{\sqrt{x^2/n}} = \frac{x}{\sqrt{x/n}}$$

(6) Log Normal Distribution:

$$f(x; \mu, \sigma) = f(x) = \frac{1}{x\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2}$$

(7) Exponential Distribution:

$$f(x) = \lambda e^{-\lambda x} \text{ OR } f(x) = \frac{1}{\lambda} e^{-x/\lambda}$$

(8) Gamma Distribution:

$$f(x) = \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} e^{-x/\beta}$$

$$\alpha > 0$$

OR

$$\beta > 0$$

$$f(x) = \frac{1}{\Gamma(\alpha+1) \beta^{\alpha+1}} x^\alpha e^{-x/\beta}$$

$$\Gamma(\alpha) > 0$$

$$\Gamma(\alpha+1) > 0$$

(9) Beta Distribution 1st kind.

$$f(x) = f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$\alpha, \beta > 0$$

OR

$$f(x) = f(x; \alpha+1, \beta+1) = \frac{1}{B(\alpha+1, \beta+1)} x^\alpha (1-x)^\beta$$

(10) Weibull Distribution:

$$f(x) = ab x^{b-1} e^{-ax^b} \quad [a, b > 0]$$

(11) chi-square Dist:

$$dF(x^2) = \frac{1}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-x^2/2} (x^2)^{n/2-1} dx^2$$

(12) F-Distribution:

$$F = \frac{x_1^2/n_1}{x_2^2/n_2}$$

(13) pareto Distribution

$$f(x; K, \alpha) = \frac{\alpha K^\alpha}{x^{\alpha+1}}$$

(14) Beta Distribution 2nd kind:

$$= \frac{1}{\beta(\alpha, \beta)} \frac{x^{\alpha-1}}{(1+x)^{\alpha+\beta}}$$

(15) Order statistics:

$$g(y_r) = \frac{n!}{(r-1)!(n-r)!} (F(y_r))^{r-1} (1-F(y_r))^{n-r} f(y_r)$$

Smallest observation: $r=1$

largest observation: $r=n$

$$f(x) = \frac{2}{2^{n/2} \sqrt{n/2}} e^{-x^2/2} (x)^{n-1}$$